

Homework (Habanero)**Q1.** Evaluate the expression: $2 \cdot 3 + 12 \div 4$

- (A) 9
- (B) 10
- (C) 11
- (D) 12
- (E) 13

Q2. Simplify the expression: $5x + 2x - 3x$

- (A) $4x$
- (B) $5x$
- (C) $3x$
- (D) $2x$
- (E) x

Q3. A patient is given $2 \cdot (5 - x)$ milligrams of medicine. If $x = 2$, how many milligrams are given?

- (A) 6
- (B) 8
- (C) 10
- (D) 12
- (E) 14

Q4. The number of cells in a Petri dish triples every 2 hours. If there are 12 cells initially, how many will there be after 4 hours?

- (A) 36
- (B) 54
- (C) 72
- (D) 108
- (E) 144

Q5. Simplify the expression: $(12 - 8) \div 2 + 5$

- (A) 5
- (B) 6
- (C) 7
- (D) 8
- (E) 9

Q6. A nurse administers $3 \cdot (x + 2)$ units of insulin. If $x = 4$, how many units are administered?

- (A) 12
- (B) 15
- (C) 18
- (D) 21
- (E) 24

Q7. Evaluate the expression: $16 \div 2 + 11 - 3$

- (A) 13
- (B) 14
- (C) 15
- (D) 16
- (E) 17

Q8. The length of a hospital room is $2x + 5$ feet and the width is $x - 2$ feet. If $x = 7$, what is the area of the room?

- (A) 50
- (B) 60
- (C) 70
- (D) 80
- (E) 90

Open Ended Questions (FRQ)**Q9.** A pharmaceutical company produces $2x^2 + 5x - 3$ units of medicine per day. If x represents the number of hours the factory operates, find the total units produced in a 5-hour period. Show all work and simplify your answer.**Q10.** A doctor prescribes $x + 2$ milligrams of a medication to a patient. If the patient's body absorbs $3 \cdot (x - 1)$ milligrams, what is the net amount of medication in the patient's system after 2 hours if $x = 5$? Show all work and simplify your answer.**Chef's Tips**

- Read each question carefully before answering
- Use the correct order of operations when evaluating expressions
- Check your work and simplify your answers